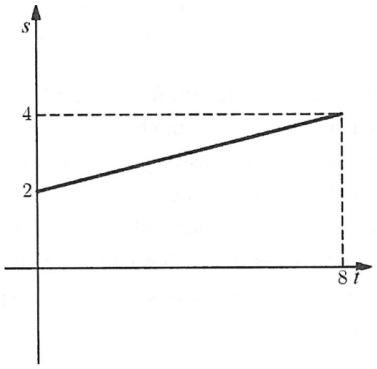
Graphs of Motion

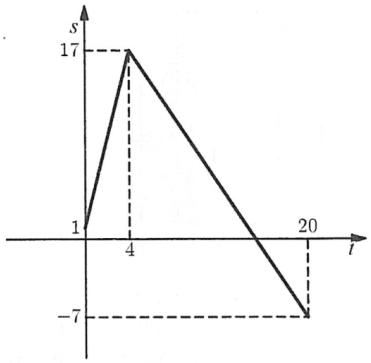
Questions 1–24

Question 1. A battery-powered toy car is placed on a straight track. The car is switched on, so that when it contacts the track, it instantly moves with constant velocity. The following graph shows the displacement s (metres), over time t (seconds) of the toy car:



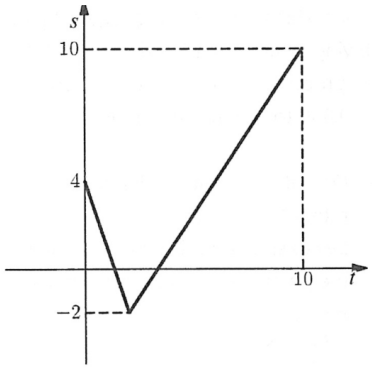
a) How far along the track is the car initially placed?
b) How far along the track is the car when 8 seconds have passed?
c) Find the velocity of the car.
d) How far along the track is the car after 5 seconds?

Question 2. Displacement s (metres) over time t (seconds) for a particle is given in the graph below:



Find
a) the distance travelled over the first 20 seconds.
b) the displacement over the first 20 seconds.
c) the average speed of the particle.
d) the velocity for the first part of the motion.
e) the velocity for the second part of the motion.
f) the average velocity of the particle.

Question 3. Displacement s (metres) over time t (seconds) for a particle is given in the graph below:



Find, after the first 10 seconds,

a) the distance travelled by the particle.
b) the displacement of the particle.
c) the average speed of the particle.
d) the average velocity of the particle.

Question 4. Lisa runs in a straight line for 60 seconds at a constant speed of 3 ms^{-1}. She then pauses for 20 seconds before turning and running in the opposite direction for 45 seconds at 4 ms^{-1}. Draw a displacement-time graph to show the athlete's motion.

Question 5. A child is running along a straight, flat path. He runs for 6 seconds at 2 ms^{-1}. He then stops and waits for 5 seconds, before running back along the path for 9 seconds, returning to his original starting point.

a) Sketch the displacement-time graph for the motion of the child.

b) Find the exact velocity for the final stage of the motion.

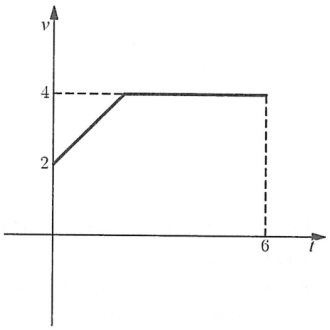
Question 6. An object travels 20 metres at constant velocity v ms^{-1} taking t seconds. It then pauses and remains stationary for 5 seconds before returning to its original position in $\frac{1}{2}t$ seconds.

a) Sketch a displacement-time graph for the motion of the object.

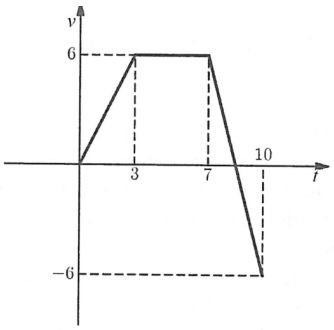
b) Given the total time the object is in motion is 6 seconds, find the value of t

Question 7. Describe what is happening to the objects in each of the following velocity-time graphs for velocity v, in metres per second and time t, in seconds. For each stage of motion comment on the speed and where possible its position:

a)

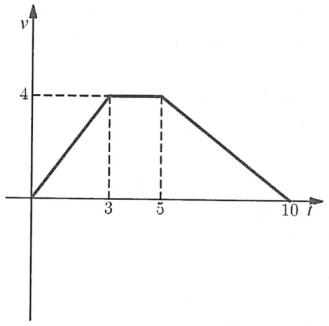


b)

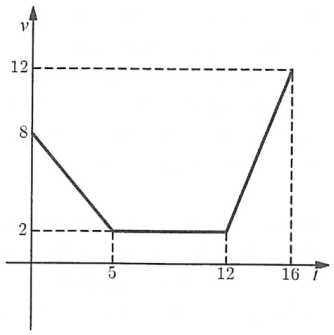


Question 8. For each of the following velocity-time graphs below, sketch the corresponding acceleration-time graph:

a)

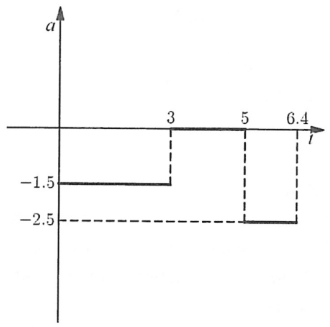


b)

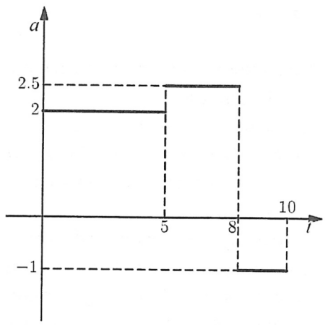


Question 9. For each of the following acceleration-time graphs below, sketch the corresponding velocity-time graph given that the object is initially at rest:

a)



b)

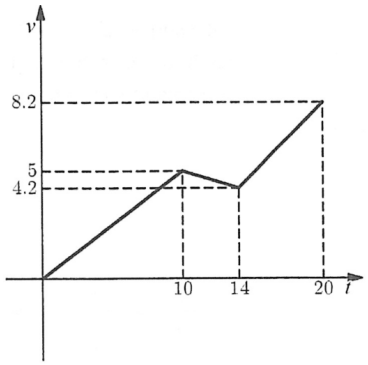


Question 10. A car which is initially at rest, accelerates at a constant rate of 4.5 ms^{-2} for 3 seconds and then remains at this constant speed.

a) Sketch a velocity-time graph to show the first 10 seconds of the motion of the car.

b) Find the distance travelled by the car in the first 10 seconds of motion.

Question 11. A cyclist is timed along a section of track, with velocity v recorded in metres per second at $t = 10$, $t = 14$ and $t = 20$ seconds.
The velocity-time graph for the cyclist is shown below.



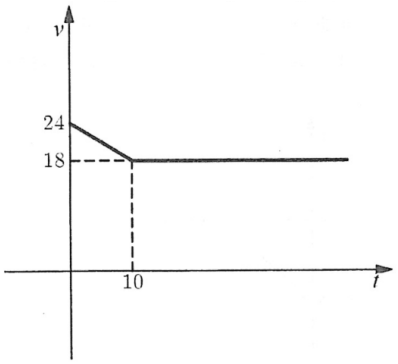
Find the

a) acceleration of the cyclist at each stage.
b) total distance travelled by the cyclist.
c) time when the cyclist is travelling at exactly 4 ms^{-1}.
d) speed the cyclist is travelling at 17 seconds.

Question 12. Sketch a velocity-time graph which represents a car initially at rest, with:

a) constant acceleration for the first 6 seconds before reaching a fixed velocity of 10 ms^{-1} which it remains at for a further T seconds.
b) increasing acceleration for the first 2 seconds, constant acceleration for the next 3 seconds, then decreasing acceleration for the next 1 second, before reaching a fixed velocity of 10 ms^{-1} which it remains at for a further T seconds.

Question 13. The velocity v ms^{-1} at a given time t seconds of a van in motion on a straight flat road is modelled using the velocity-time graph below:



a) Find the deceleration of the van in the first 10 seconds.
b) At 10 seconds, the van continues to travel at a constant speed of 18 ms^{-1} for a distance of 1026 m. Find the time it spends at this speed.
c) Find the total distance travelled by the van at the time given in your part b).

Question 14. An object is moving in a straight line. It starts at 4 ms^{-1}, before accelerating uniformly to 10 ms^{-1} in 8 seconds, moving at 10 ms^{-1} for 5 seconds before finally decelerating to rest in 7 seconds.
a) Sketch a velocity-time graph which represents the motion of the object.
b) Find the distance travelled by the object.
c) Find the average velocity of the object.

Question 15. A van is driving along a straight stretch of motorway. It starts at 33 ms^{-1}, before decelerating to 22 ms^{-1} uniformly over 1.1 km.

a) Sketch a velocity-time graph which represents the motion of the van.

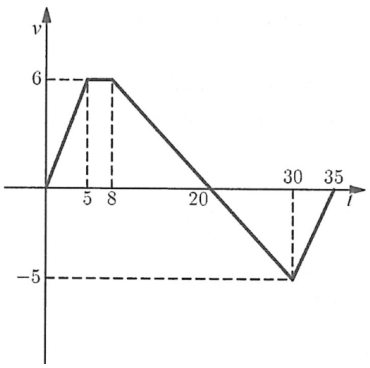
b) Find the time the van spends decelerating according to this model.

The van then maintains this speed for a further 400 metres.

c) Find the average speed of the van.

d) Two speed cameras, one at the start and one at the end of this 1.5 km stretch of road, measure the speed of the van and finds its average. Find this speed.

Question 16. A vehicle is being driven along a flat, straight road. Its motion is modelled using the velocity-time graph below where velocity v is given in metres per second and time t is given in seconds.



a) Find the acceleration at each stage of motion.

b) Find the total distance driven by the vehicle in the first 20 seconds.

c) Find the total distance driven by the vehicle in the final 15 seconds.

d) State the displacement of the vehicle at $t = 20$ seconds.

e) State the displacement of the vehicle at $t = 35$ seconds.

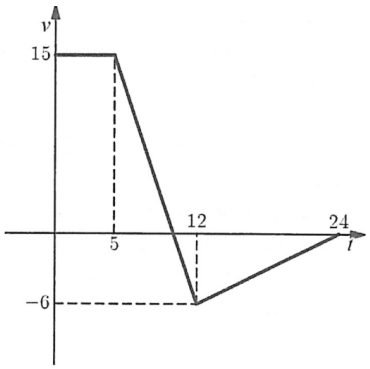
f) State the total distance driven by the vehicle.

g) Find the average speed of the vehicle over the 35 seconds.

Question 17. A cyclist starts from rest and is cycling along a flat straight road. He accelerates at a constant acceleration of 0.5 ms^{-2} for the first 5 seconds, before accelerating at a constant acceleration of 0.3 ms^{-2} for the next 5 seconds. He then remains at a constant speed for a further 20 m.

a) Sketch a velocity-time graph to model the motion of the bicycle.
b) State the velocity of the cyclist in the third stage of the motion.
c) Find how far the cyclist has cycled since the initial start point.

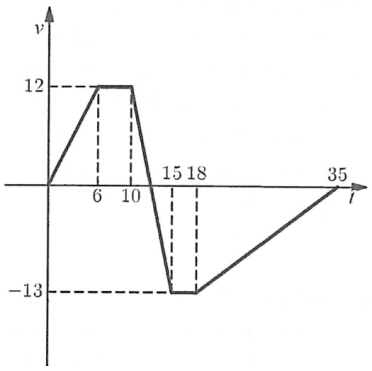
Question 18. The velocity v ms^{-1} at a given time t seconds of a car in motion is modelled using the velocity-time graph below.



Find

a) the acceleration for all three stages of the motion.
b) the exact times at which the car, according to the graph has instantaneously stopped.
c) the total distance travelled by the car.
d) the displacement of the car from its initial position.

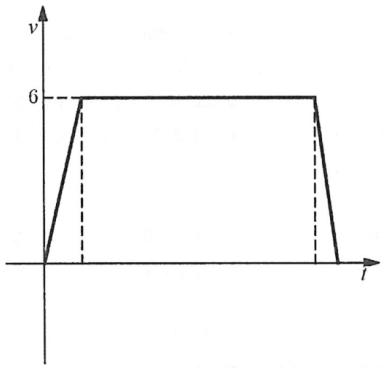
Question 19. A ride at a theme park with a 35 second duration has a carriage that travels along a flat, straight track.
The velocity-time graph below shows the motion of the carriage in time t in seconds.



a) Find the maximum magnitude of acceleration over the 35 second ride.
b) Find the times at which the carriage is instantaneously stationary.
c) If the passengers disembark at 35 seconds, where are they in relation to the start point?

Question 20. A bus travels from rest from a bus stop, on a straight, flat road. It drives with acceleration 0.9 ms^{-2} for 15 seconds. It is then driven at constant velocity before decelerating uniformly at 1 ms^{-2} where it comes to rest at a second bus stop which is 918 metres from the first bus stop.
a) Find the velocity of the bus when it is moving with constant velocity.
b) Sketch a velocity-time graph to represent the motion of the bus.
c) Find the time taken to get between the bus stops.

Question 21. The velocity-time graph shows the motion of a runner running along a straight flat track of length 200 metres before coming to a stop.



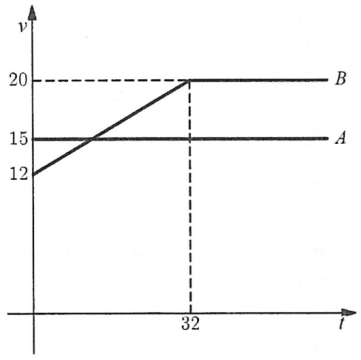
The runner accelerates uniformly from rest over the first 15 metres until reaching a speed of 6 ms^{-1} before continuing to run at 6 ms^{-1} until they complete the 200 m. A further 3 seconds later they come to rest. Find

a) the exact time the runner completes the 200 metres.
b) the total distance they run.

Question 22. An object starts from rest and is travelling in a straight line with uniform acceleration for T seconds, travelling 24 metres. It then moves with constant velocity for 9 seconds before coming to rest uniformly in a further $\frac{3}{5}T$ seconds.

a) Sketch a velocity-time graph for the motion of the object.
b) Given the displacement of the object is 146.4 metres, find the value of T

Question 23. Two cars, A and B are driving along a straight, flat road.
Car A is driving with constant velocity 15 ms^{-1}. Car B, which is initially driving at 12 ms^{-1}, is driving with constant acceleration until 32 seconds have passed at which it then drives at 20 ms^{-1}.
The velocity-time graph demonstrating the motion of the two cars is shown below:



a) Find the acceleration of car B for the first 32 seconds of the motion.
b) Find the equation that gives the velocity of car B for the first 32 seconds of motion.
c) Find the exact time at which car B overtakes car A according to the model.
d) Find the distance the cars have travelled when car B overtakes car A.
e) State what the point of intersection of the two lines represents.

Question 24. Two cars, A and B are driving along a straight, flat road. Car A is driving with constant velocity u ms^{-1}. Car B, which is initially driving at $(u-5)$ ms^{-1}, is driving with constant acceleration until T seconds have passed at which it then drives at a constant velocity. Car B overtakes car A at exactly T seconds.

a) Sketch the velocity-time graph for the two cars.
b) State the velocity of car B when it overtakes car A.
c) Given that the cars are going the same speed exactly when $t = 20$ and car B overtakes car A after driving 1000 m, find the values of T and u